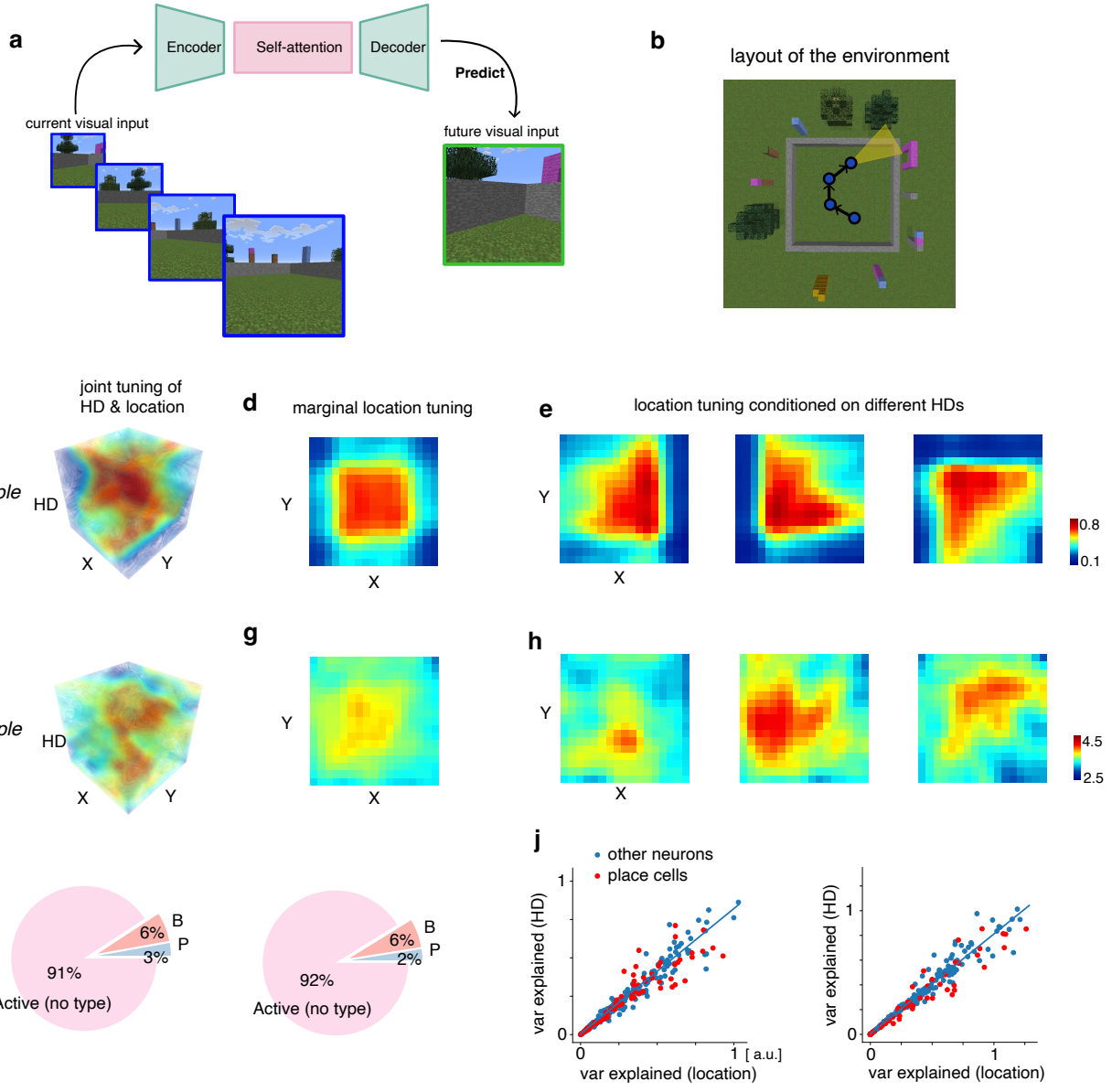

a
Encoder
Self-attention
Decoder
Predict
current visual input
future visual input
b
layout of the environment
c
joint tuning of HD & location
example unit 1
HD
X
Y
d
marginal location tuning
Y
X
e
location tuning conditioned on different HDs
Y
X
0.8
0.1
f
example unit 2
HD
X
Y
g
Y
X
h
Y
X
4.5
2.5
i
6%
B
3%
P
91%
Active (no type)
6%
B
2%
P
92%
Active (no type)
j
other neurons
place cells
var explained (HD)
var explained (location)
1
0
[a.u.]
var explained (HD)
var explained (location)
1
0
1